

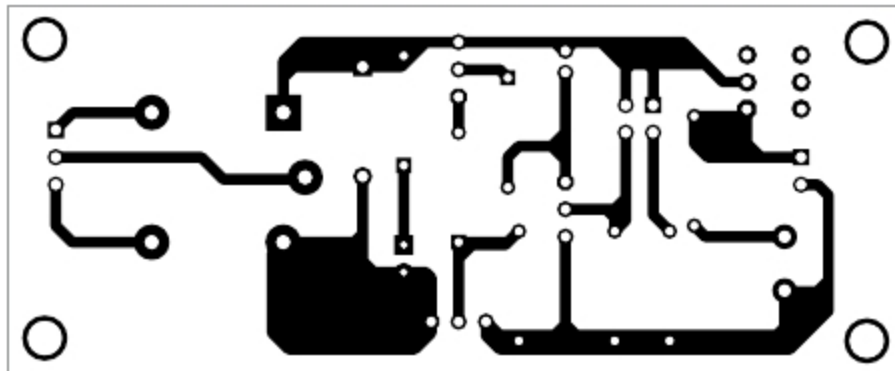


Fig. 3: Actual-size PCB layout of automatic IR faucet controller

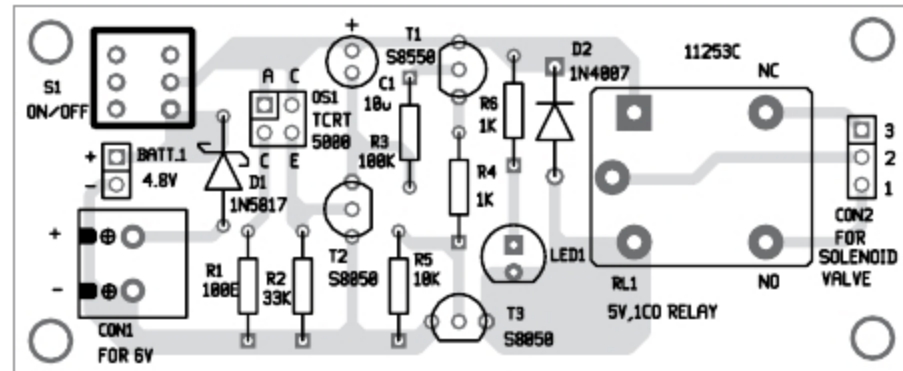


Fig. 4: Components layout for the PCB

Construction and testing

An actual-size PCB layout of the automatic IR faucet controller circuit is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on the PCB, connect 6V across CON1.

Build the circuit with the battery pack in a waterproof enclosure. Make sure that the IR sensor can easily detect hands near the washbasin.

Note that, selection and installation of the solenoid valve depends on specific requirements. If you have a low-voltage (6V) solenoid valve, it may work with the 4.8V battery pack (most 6V types work smoothly with 4.8V).

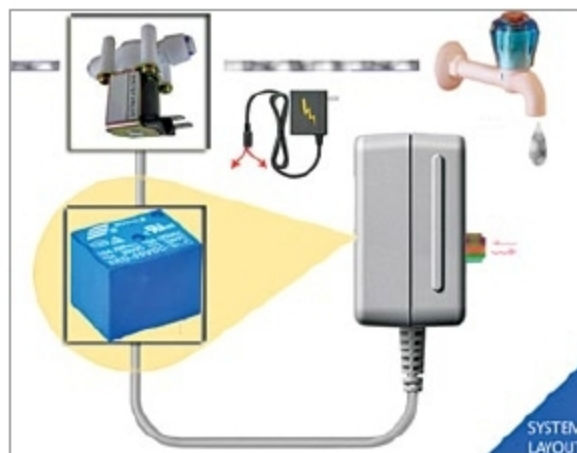


Fig. 5: Proposed faucet arrangement

But if you have a high-voltage valve (greater than 6V), connect it to an external power supply. Wiring diagram of solenoid valve is shown in Fig. 6.

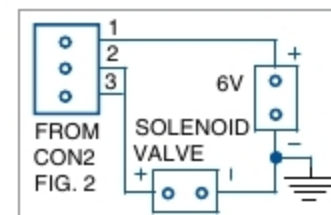


Fig. 6: Wiring diagram of solenoid valve

A proposed faucet arrangement of the system is shown in Fig. 5. The circuit draws under 50mA current in quiescent state. It goes above 100mA in active state if a low-voltage solenoid valve is also powered by the circuit. **EFY**



T.K. Hareendran is founder and promoter of TechNode Protolabz

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